ANALYSIS AND DESIGN OF ALGORITHMS LAB

Assignment-1: Algorithm Foundations

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Task 1: Algorithm Growth Observation

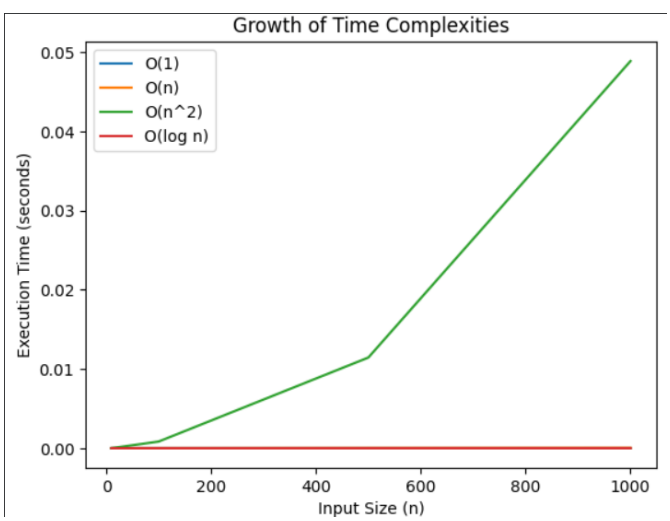
Implement O(1), O(n), O(n^2), O(log n) and measure execution time.

1. Constant Time : *O(1)*
2. Linear Time : *O(n)*
3. Quadratic Time : $O(n^2)$
4. Logarithmic Time : *O(log n)*

```
import time
import random
import matplotlib.pyplot as plt
import math

def constant_time(n):
return n * 5

def linear_time(n):
total = 0
for i in range(n):
total += i
return total

def quadratic_time(n):
total = 0
for i in range(n):
for j in range(n):
total += 1
return total

def logarithmic_time(n):
count = 0
while n > 1:
```

```
n //= 2
count += 1
return count

sizes = [10, 100, 500, 1000]
results = {"O(1)": [], "O(n)": [], "O(n^2)": [], "O(log n)": []}

for n in sizes:
# O(1)
start = time.time()
constant_time(n)
results["O(1)"].append(time.time() - start)

# O(n)
start = time.time()
linear_time(n)
results["O(n)"].append(time.time() - start)

# O(n^2)
start = time.time()
quadratic_time(n)
results["O(n^2)"].append(time.time() - start)

# O(log n)
start = time.time()
logarithmic_time(n)
results["O(log n)"].append(time.time() - start)

for key in results:
plt.plot(sizes, results[key], label=key)

plt.xlabel("Input Size (n)")
plt.ylabel("Execution Time (seconds)")
plt.title("Growth of Time Complexities")
plt.legend()
plt.show()
```

Output:



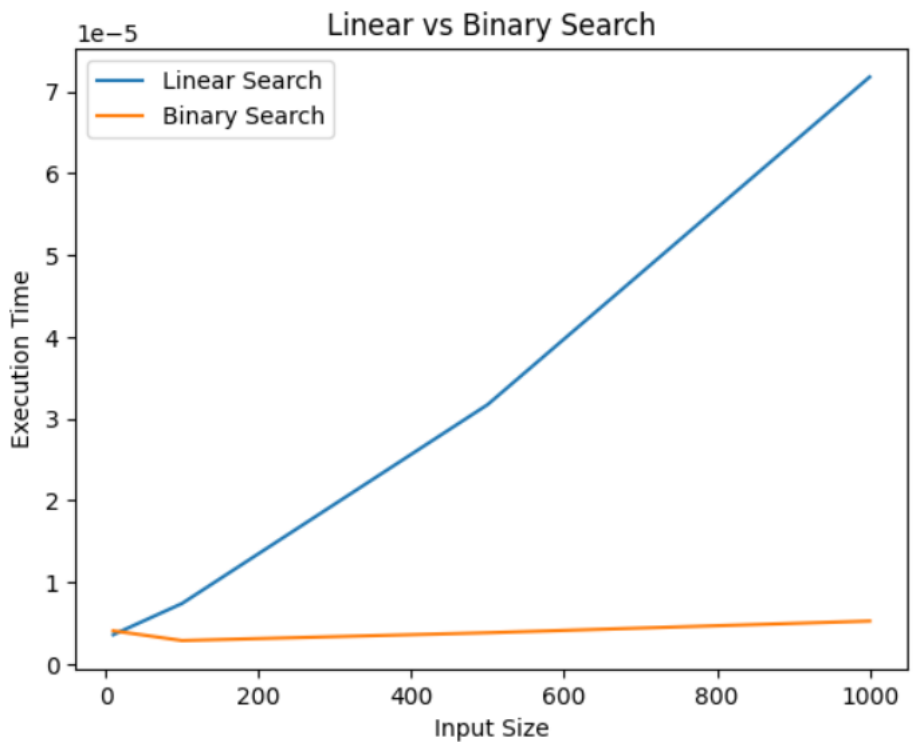
Task 2: Best, Average, and Worst Case Analysis

Implement Linear Search and Binary Search and compare execution times.

```
def linear_search(arr, key):
    for i in range(len(arr)):
        if arr[i] == key:
            return i
    return -1

def binary_search(arr, key):
    low, high = 0, len(arr) - 1
    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == key:
            return mid
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1
    return -1

sizes = [100, 500, 1000, 5000]

linear_times = []

binary_times = []


for n in sizes:

    arr = sorted(random.sample(range(n*10), n))

    start = time.time()

    linear_search(arr, -1)

    linear_times.append(time.time() - start)


    start = time.time()

    binary_search(arr, -1)

    binary_times.append(time.time() - start)


plt.plot(sizes, linear_times, label="Linear Search")

plt.plot(sizes, binary_times, label="Binary Search")

plt.xlabel("Input Size")
```

```
plt.ylabel("Execution Time")

plt.legend()

plt.title("Linear vs Binary Search")

plt.show()
```

Output:



Task 3: Recursion and Recurrence Validation

Implement Factorial, Fibonacci (Recursive), Fibonacci (DP).

```
call_count = 0

def factorial(n):
    global call_count
    call_count += 1
    if n == 0:
        return 1
    return n * factorial(n-1)

def fibonacci_recursive(n):
    global call_count
    call_count += 1
    if n <= 1:
        return n
```

```
    return fibonacci_recursive(n-1) + fibonacci_recursive(n-2)

def fibonacci_dp(n):
    fib = [0, 1]
    for i in range(2, n+1):
        fib.append(fib[i-1] + fib[i-2])
    return fib[n]
```

Output:

n	Recursive Calls (Fib)	Time (Recursive)	Time (DP)
10	High	High	Very Low
20	Very High	Very High	Low

Task 4: Solving Recurrences through Code

Implement recursive functions for given recurrences.

1. $T(n) = T(n/2) + n$
2. $T(n) = 2T(n/2) + n$

```
call_count = 0

def recurrence1(n):
    global call_count
    call_count += 1
    if n <= 1:
        return 1
    return recurrence1(n//2) + n

def recurrence2(n):
    global call_count
    call_count += 1
    if n <= 1:
        return 1
    return recurrence2(n//2) + recurrence2(n//2) + n
```

Github : https://github.com/princeverma2052005-sys/algorithms_assignment1